DCLAB · R&D CONVERSATION

Your questions, from the first to the latest

Every message you typed in this chat, word for word and in order, with a short note on what came back for each.

43	9	19 Sep 2026
questions	topics	to 04 Oct 2026

Topics

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How this was made. Questions are copied exactly from the saved session transcript, including original spelling. Times are Dubai time. System notices and automatic messages are left out, and one question you sent twice appears once. The “What came back” notes are short summaries; the full detail is in *DCLab_Chat_Recap.md*.

01 R&D data and fine-tuning data

Q1 19 Sep 2026 · 16:24

i made below project and i want know how can extract more cleaner and clear and simpler data from this information and also make the data more cleaner for fine-tuning the SLM for better data science and machine learning enginering
now in this R and D project i want make 50 experiment and on real dataset in kaggle github or any hugging face or any good resources we can download and run as much as we can experiment and get huge knowlege and context of the process and procedure of creating models in first steps even that is data analyses eda and understand and deep understanding the in the first step the data processing transformation feature engineering which feature are lekage which one are not how it efect how we can generate more features that are not exist in the cols of data or eve the data and then collect all these context of feature engineering and even best mothod how we can understand which feature are reliable or maybe they miss in the production best way that we can understand with feature are really leakag features and consider in all steps that you can acccess best LLM agent to help you which is very much important and the tarinig model whic algorithm which model works best and the how we can optimized the model training the which methods and strategy works best for which kind of data which models which algorithm are important
again consider in all steps that you can acccess best LLM agent to help you
the final goals are first getting huge knowlege and learn from these huge experiments both theorical and practical know that a LLM agent can get from these experiment ,
the next goal is the LLM agent can understand best model best algorithms best feature engineering and best trianing , best optimiztion
and next goal is getting huge conetect of codes not to make lines of code the flow of codes or blocks of code that realy work fine
the final goal is understanding the process of procedure and workflow of buiding the best model that we can and then collect these prcesses of workflow and procedure of building models for trainnig LLM for future
now satrt to do it and automatate all these goals

i want make very better and complete review that even a person from out side read it and understand it and the help to understand more and make very ituition and undertanding and guide and give recommendations or suggetion and find rule and laws for future use
caseshttps://github.com/Shahriyar-Moradi/Research_Development_DCLab/tree/main
https://github.com/Shahriyar-Moradi/Research_Development_DCLab/tree/main
https://github.com/Shahriyar-Moradi/Research_Development_DCLab

Sent with the document Decision_AI_Agent_Coding_Context.docx.

WHAT CAME BACK

Found the 50-experiment campaign already complete (10 UCI datasets × 5 stages, 22 rules, 10 workflow blocks). Delivered **DCLab_RD_Review_and_Roadmap.md** and **build_sft_dataset.py** (104 clean SFT examples).

Q2 19 Sep 2026 · 17:41

> ok now fill the gaps for me
> and add Kaggle/HuggingFac and then add multiclass, time-series, imbalanced-fraud, or text+tabular cases yet.
> ok how can i make paired instructionand then how can i add SFT job can train on directly?
> add single outside-reader-friendly narrative doc distinct from the (excellent but dense) field guide.
> i do not want remove any current thing i want add beside it better context for how can extract more cleaner and clear and simpler data from this information and also make the data more cleaner for fine-tuning the SLM for better data science and machine learning enginering
> the goal is improvement
> the final goal of the the right and correct and automated R&D for DClab project for make the best mvp

WHAT CAME BACK

Added **dataset_expansion_pack.py** (fraud, multiclass, time-series, text+tabular Kaggle adapters), **build_sft_dataset_v2.py** (real 5-part reasoning traces), **SFT_DATA_GUIDE.md** (JSONL formats + LoRA script) and **DCLab_Field_Notes_Plain_Language.md**. Nothing removed.

Q3 19 Sep 2026 · 22:49

> but in all example you mention the experiment what the LLM or SLM does not know about the experiment what was is about how it created a or other knowladge
> this is one the example: {"messages": [{"role": "system", "content": "You are a senior data-science / ML engin
> eering research assistant. You answer using the DCLab evidence-first model-building f
> ramework: state the observed evidence, your interpretation and its uncertainty, a con
> crete decision or recommendation, remaining risks or missing evidence, and the smalle
> st next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-023
> on the 'credit_default' dataset find during the 'feature engineering' stage?"}, {"rol
> e": "assistant", "content": "Other findings:\n- Use the raw feature recipe for the
> xt model screen on credit_default; it achieved 0.7317 mean training-CV ROC-AUC with a
> bout 23 features.\n\nKnown limitations:\n- Generic mathematical feature generation ma
> y lack domain meaning; a domain expert must review production semantics."}], "metadat
> a": {"source": "agent_memory", "experiment_id": "EXP-023", "dataset": "credit_defaul
> t", "stage": "feature_engineering"}}

WHAT CAME BACK

You were right: examples named an experiment ID without context. **build_sft_dataset_v3.py** adds a dataset card, stage card and setup summary to every example.

02 Using the data: RAG, tools and the DCLab product

Q4 19 Sep 2026 · 22:57

> and also how can i use these data with fine tuning that dclab agent can use this rules to make this workflow machine learning and data science use cases and flow the steps
> I know one the methods is RAG but How must be data for RAG
> and also is there other techniques to do it or any other methods what can work fine even better than a RAG or is there other methods like make it agentic or tools?

WHAT CAME BACK

RAG, fine-tuning and tools each do a different job. RAG should filter by dataset/stage first, then rank. Delivered **build_rag_index.py** and **AGENT_ARCHITECTURE_RAG_TOOLS_FINETUNE.md**.

Q5 20 Sep 2026 · 08:38

> now how can i use it practically for now without finetuning or use RAFT
> now guide to use my dclab
> how can I improve it to use better
> guide me

WHAT CAME BACK

A no-training ladder: retrieval before prompts, then **grounded_review.py**, then **dclab_knowledge_mcp_server.py** (3 tools tested on real data), then tool calls inside LangGraph.

Q6 20 Sep 2026 · 08:58

> ok now guide me to implement it how can use all of these data from experiments in useful way for dclab how can use evry details that i generated every content and context and experiment details to use practically and then implement this is what i want

WHAT CAME BACK

Read the real DCLab product code. Its spec forbids a parallel agent framework, so **DCLab_Product_Integration_STEPS.md** feeds R&D evidence into the existing leakage-review checkpoint (new field, v2 prompt, schema enum, ledger).

Q7 20 Sep 2026 · 09:12

> but the most important thing of R&D project is behinde the sense to just help the dclab make the paln to how i can take advantage of all of data of the r and d it for dclab for now and next phase i want use it as R&D part in dclab

WHAT CAME BACK

DCLab_RD_Operating_Model.md: R&D as a standing function that earns claims before they ship, with a loop that closes both ways and findings built into pipeline defaults.

Q8 20 Sep 2026 · 09:55

> you understand it more better now i want build something reliable with evidence that every data scientist and machine learning engineer exactly i want to fill gap do colab anad notebooks in the agentic way and make these workflow automate and then do some r and d to verify my job is implementing correctly and then give this r and d tool to make their model in best way i want and dclab really help them

Sent with an attached vision document (“Cursor for Machine Learning”).

WHAT CAME BACK

Most of the vision was already backed by evidence; the risky new piece is the autonomous loop. **DCLab_Agentic_Loop_RD_Verification_Plan.md**: replay, red-team, constrained moves, logged iterations.

03 Notebook copilot and UI/UX design

Q9 20 Sep 2026 · 10:06

> do know i want intellegent agent that can correct and help the data scientist and machine learning engineers write comment or suggestion beside evry cells of code in the notebook or help make their model faster and effiecent and optimized which feel statisfied show we can make them more than be productive be productive with correct and tested evidence with tested experiments that we did the data scientist and machine learning engineers they
> trust us something like 50 kaggle challenges soleved and help you realtime beside you or in the chatbot can ask to show the proof and make you sure see the similar examples are really helpful something like stackoverflow that we used so many years ago that we saw some body else in the world face this issue and solve it and it is the proof of solution and and data scientist and machine learning engineers can use it

WHAT CAME BACK

Working prototype **notebook_copilot.py** with 4 detectors, each citing a real rule and precedent, plus **NOTEBOOK_COPILOT_SPEC.md** (including an honest look at the “50 Kaggle challenges” claim).

Q10 20 Sep 2026 · 14:19

> how can you design it i mean UI and UX of it

WHAT CAME BACK

First inline mockup: the warning sits under the exact line, with a short “Show proof” panel and Apply fix / Dismiss / Ask follow-up.

Q11 20 Sep 2026 · 14:32

> can you design it better even write comment or suggestion beside evry cells of code in the notebook or help make their model faster and effiecent and optimized
> or be more creative also design it with more better graphic

Sent with reference screenshots.

WHAT CAME BACK

Numbered step-flow mockup: scan → findings → model comparison → bounded tuning.

Q12 20 Sep 2026 · 19:54

> Why was LightGBM selected over XGBoost and random forest for this notebook, with evidence?

WHAT CAME BACK

Corrected the mockup’s invented numbers with real EXP-004 data: LightGBM beat hist_gradient_boosting by only 0.0002, decided on stability and speed.

Q13 21 Sep 2026 · 09:01

> ok continue

WHAT CAME BACK

Ran a real 5-model screen on synthetic churn data: **logistic regression won** (0.7757 vs LightGBM 0.7313). Saved as an experiment record.

Q14 21 Sep 2026 · 09:07

> use 2 attached image to better UI and UX not same as attached images make better and improve them to have better colab and notebook with agentic functionality and much better UI

Sent with 2 reference images.

WHAT CAME BACK

Called out problems in the reference images (hype copy, invented numbers, a “Looks good!” banner beside open risks) and redesigned with real numbers.

Q15 21 Sep 2026 · 09:11

> now design a new Colab with agentic AI functionalty entire web pages with or even better more creative to redesign the colab that works with agentic AI

WHAT CAME BACK

A full “dclab notebook” page: cells plus agent panel, optimization trail shown in context.

Q16 21 Sep 2026 · 09:16

> yes mock up more better with web page n a different visual direction entirely with better graphical UI

WHAT CAME BACK

A different direction: the page is the ML pipeline graph, with a command bar instead of a chat panel.

Q17 21 Sep 2026 · 09:18

> I attached images use them to design better a webpage even in the html

Sent with reference images.

WHAT CAME BACK

Built and published **dclab_studio.html** as a live page.

04 A focused agentic model

Q18 21 Sep 2026 · 14:00

> ok let's put the design away now
> how can build model LLM or SLM model i mean fine tune like a qwne 3.8 27B or even a model like 8B
> how we can do it to act as agentic instead making agentic system around the model the model itself act as agentic in my case, the model knows the what is next step and is the procudure and what is the workflow and should to do next step of building a machine learning model in my case
> something better than the LLM-instruct models which know the workflow understand each block of machine learning and machine learning buiding
> in general how it can be possible ?
> what i want to in second question how we can use these inscruction and workflow and procedure as token not like sub words like normall LLM, I want something bigger as token
> how it will be feasible ?

WHAT CAME BACK

A 4-rung ladder to make the model itself agentic (declarative SFT → trajectory SFT → rejection sampling with the validator → RL), plus action/macro tokens with code.

Q19 21 Sep 2026 · 14:43

> i just made an example of qwen models since they perfome better in opensources i just want clarify it as example for you to understand you guide me what should i do to get this approach
> let's think this way or mabye generate 2 steps togther instead of generating separately
> for example for writing function instead of each word for write lines and each lines will be a function making LLM model that can generate a entire function in a token or each lines of funtion as a token instead of words as tokens
> or i think something like cooking a cake steps in normal way: step1: eggs 2 add suger , step 3 mix these togther 4, add milk then mix again
> my way is step 1 an 2 both generate as one suger and egss mixed togther and ready to add milk instead of breaking merge them asleast at the generation layer of LLM

WHAT CAME BACK

Your “cake” idea split in two: fusing steps in the training data (buildable now, and a safety win) vs. multi-token prediction in the architecture (later).

Q20 22 Sep 2026 · 00:46

let me ask in this way it good way of think or idea to make this at all? first my DClab then expand for other workflows not only machine learning and data science workflow? in other industrries? look my idea critically? maybe my way of the thinking is totaly wrong?
what i want is improve the understand the workflow and the final goal for now only internal workflow with not huge steps , for example 10 steps fair enough
in this R and D project the goal is to make not only the data science and machine learning experiment but also have workspace to test orthing and R and D to make better development product pratically not only in the chat and the codes after the coding i want to see and test another project in the R and D lab and stadio can this agentic llms
there are some reason also to build this model if its feasible again which reduce hallucination and myabe instead of making agent with llm with lot of tools and harnesses and other thing to make it determinecti more the model can act determinictic by itself and the final goal of this model is fully concentrate and focus not distracted that i really want buid literly a model focused and concentrate one to utimatley 10 workflws with their details of the worflow or wrokflow that is final goal that i want to achive more thing, i dont want make gpt astra or cluad fable not at all never, just focused and concentrate dominate on details of workflow no more thing without forgetting details and the forgetting is the critical problems in fine tuning
another reason of building is a model can generate combined answer for 2 steps for example step 1 LLM generate 200 tokens and step 2 200 tookens individually but i want generte answer or do the tasks by this method instead of genearting 400 token for do the task of step 1 and 2 and answer these 2 steps question only generate 200 token totally that covers the both steps answer and do the both task as agent with only on single time token generation and then extend this for 5 steps or 5 tasks with only generating token for a task
another thing is i want feasible innovation which is also practical and the important thing you guide me
understand it how this idea can be feasible and if it's let implement and test then if it's work develop it practically and then use it practically in dclab product
the if we this idea works we can r and for another if we can made agentic llm or slm can we transfer knowladge of it to another things like making better perdictive models with higher percision and recall that model can learn the historical data with tranformers model or any other type of the models
that's why i build R and D lab stadio

WHAT CAME BACK

Honest critique: the narrow 10-workflow idea is sound, but fine-tuning alone isn’t deterministic, expansion depends on having a validator, and token compression has a floor.

05 Token and cost efficiency: NOOA and Jev

Q21 22 Sep 2026 · 08:33

> how can make it effiecent and with minimum tokens? for example nvidia make nooa nvidia object oriented agent to make less token consumer
> now how can i engineer this system to decrease amount of token consumption

WHAT CAME BACK

How NOOA saves tokens (pass by reference, fewer calls). Found an unbounded `_evidence_json()` in the product and proposed a bounded preview.

Q22 22 Sep 2026 · 08:45

> what about new tool typesafe Jev ?
> how can i be innovative to make better model ? more effiecent

WHAT CAME BACK

Jev as “System 1” for typed decisions, with a fine-tuned model as “System 2”; you can train your own System 1 from registry data.

Q23 22 Sep 2026 · 09:02

> do know jev think differently and make something innovative i just give you example to have other mindset of effiecency like nooa
> that's why i would like to at jev as innovation think differently from another point of view like steve jobs think out side box that why i told you build agentic llm or workflow llm, same like nooa that use only principal of solid coding and make agetic system effiecent and jev use again some other software coding like boolean to make decision
> and here i am want to be innovative not just a simple fine tuning workflows and pipelines

WHAT CAME BACK

The outside-the-box borrow: treat the workflow as a spreadsheet/build-system dependency graph, so the model fills one cell at a time and can’t forget.

Q24 22 Sep 2026 · 09:14

> one context you should know and update your memory that i am AI and machine learning enginner with 7 years of experince which i have enough knowlege but i ask you explain simple and make your explanation easy to understand
> now show the practical visual graph of the architecture of both model and data in good graphic anf after that we can start

WHAT CAME BACK

Memory updated (7+ years’ experience, simple explanations). Two diagrams: the data graph and the model architecture inside one step.

06 GNNs, speculative decoding and benchmarking Jev

Q25 22 Sep 2026 · 09:22

> how GNN can help in this project and this workflows

WHAT CAME BACK

Not for the 5-step workflow graph; possibly later for a knowledge graph or for picking which model to try first.

Q26 22 Sep 2026 · 17:57

> ok keep gnn away
> i watched some videos regarding specultive decoding and many method related it to like mtp , dflash, and dpark methods which opensource model are capabale of fine tuning and inferences and adoptable for these how specultive decoding and many method related it to like mtp , dflash, and dpark methods really help me?

WHAT CAME BACK

Speculative decoding (MTP, EAGLE-3, DFlash/DSpark) is a speed and cost lever, not a quality lever, and structured output is its best case. (You sent this question twice; it appears once here.)

Q27 22 Sep 2026 · 23:43

> i also research about the jev and find out it help me to select and make decision for best experiments and model selection best decision it also very useful made research regarding i think jev also can really helpful in after the fine tuning and then specultive decoding and many method related it to like mtp , dflash, and dpark methods and then use jev for decision making or other things or workflows

WHAT CAME BACK

Jev and the LLM aren’t steps in sequence: they produce different fields of the same answer at the same time. Constrained decoding on your own model is the vendor-free alternative.

Q28 23 Sep 2026 · 01:35

> but help how can i improve my workflow using jev or even the details of my workflow ? practically you my idea and goal of R and D

WHAT CAME BACK

Stage-by-stage table of which decisions are typed vs. generative, and a post-hoc agreement test using the existing registry.

Q29 23 Sep 2026 · 14:22

we can compare the model selection decsion of next step or action of thr workflow using our agentic llm and jev how much each of them are reliable?
then one more things the jev is capable to do is to classification model have you ever think about it ? it can do same as classifier model
As a data scientist, my first question is: compared to what?
We've been building classification models for decades. Here's how I see Jev:

It's a good general-purpose model.
Its appeal is zero-shot. You describe your task and get started without training your own classifier.
Want to sort customer emails into billing or technical support? That convenience is valuable.
i want add experiments for jev also and make benchmark of it comapre to machine learnng models

WHAT CAME BACK

Jev as one more row in the model screen. Delivered **jev_classifier.py** and **run_jev_screen.py**; per-row calls were ~350× slower than batch prediction even with a mock.

Q30 23 Sep 2026 · 17:30

the most important thing in usage of NOOA and Jev is optimization in the cost and token consuming since NOOA show half normal agent way consume tokens and jev also use use much less token to classify or sumerize or score or true or false cases thin in this way how we can use jev and NOOA if we have conflict NOOA and langgraph how we can fix these conlict to use best way of NOOA to derease the cost of tokens but the important thing is the performance good enough

WHAT CAME BACK

The conflict is LangGraph's growing message history. Fix: keep only references in state, route with plain code, and use a 3-tier cost ladder (typed classifier → NOOA → full LLM) with one validator.

07 Transcript and training glossary

Q31 25 Sep 2026 · 23:27

> Please compile this entire conversation, from the very first message to now, into a downloadable file — include every message in full (mine and yours), in order, with no summarizing or shortening. Save it as a [Markdown/Word/PDF] file and give me the download link.

WHAT CAME BACK

Started a verbatim transcript, caught my own mix-up, and pointed to Settings → Privacy → Export data as the reliable source.

Q32 25 Sep 2026 · 23:32

> Ok give complete fully detaol of our chat in a in pdf

WHAT CAME BACK

Restarted the transcript from the visible text; it reached Message 3 before the chat moved on.

Q33 26 Sep 2026 · 12:28

> explain most useful context of training model pharases
> like pretraining , pretraining data , ...
> explain 100 of these

WHAT CAME BACK

ML_Training_Terminology_Glossary.md: 106 terms in 6 groups, with Persian terms.

08 Other domains: computer vision, self-driving, maps, scene graphs

Q34 26 Sep 2026 · 18:00

> how can i make this workflow for other use cases?in the computer vision systems ? in autonomous driving car?

WHAT CAME BACK

Computer vision transfers almost directly. For self-driving, perception transfers but planning/control does not (no counterfactual ground truth, safety-case evidence, latency).

Q35 26 Sep 2026 · 18:03

> working on the map data to find flow and the road same like worflow

WHAT CAME BACK

Road networks are a real GNN win (Google Maps ETA). Same 5 stages, spatial leakage, a baseline before the GNN, and DeepMind’s Supersegments.

Q36 26 Sep 2026 · 18:06

> what about a agentic llm for workflow with graph structure?

WHAT CAME BACK

Three meanings; the one to build is tool calls over the graph (the agent orchestrates, the GNN predicts).

Q37 26 Sep 2026 · 18:10

> how we can Making the model's own attention mechanism graph-structured — a real research direction, not a "build this now" one it for computer vision?

WHAT CAME BACK

For ordinary images, attention is already a complete graph. Real fits: point clouds/LIDAR, scene graphs, Vision GNN.

Q38 26 Sep 2026 · 18:16

> how can start it by analys the image as complete graph
> and then Scene graphs — object relationships
> what i exaclt looking for is Scene graphs — object relationships make vision language model understand the the flow of object and then these object has relationship with each other like graph a agentic vlm understand the flow of object and then these object has relationship with each other like graph

WHAT CAME BACK

A 5-step scene-graph pipeline: detect → prune pairs → classify relationships → video for “flow” → VLM reasoning via tool calls, then test it.

Q39 26 Sep 2026 · 18:35

> how gnn work in this case simplify it and explain
> first what is use case gnn in simple use cases
> then can gnn detect these workflows and relationship even in 2d space in
> even gnn can make this flow for use or graph transformers can generate the these flows

WHAT CAME BACK

GNNs explained simply (nodes ask their neighbors), with use cases. A GNN never sees pixels; something else builds the graph first.

Q40 26 Sep 2026 · 18:43

> what is the end result of the gnn , in cnn we make classification or object detection but wht abou gnn and then what is end result of gnn in this workflow

WHAT CAME BACK

Outputs are node-, edge- or graph-level. For scene graphs: graph in, the same graph with labeled edges out.

09 Moving into the coding session

Q41 01 Oct 2026 · 14:44

> continue in code

WHAT CAME BACK

Checked for `torch/torch_geometric`; the session then moved to the coding workspace and the R&D repo was cloned.

Q42 03 Oct 2026 · 11:35

> let's recap all we have talk in a details in this chat only and generate complete markdown file

WHAT CAME BACK

DCLab_Chat_Recap.md: full recap in 13 phases with verified numbers and a Persian summary; also saved to the project.

Q43 04 Oct 2026 · 20:56

> can genearte pdf file from this chat specially my questions from beggining untile now

WHAT CAME BACK

This document.